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# Nash Equilibrium

*Part 2*

*Using Beamer*

# Different Algorithms

- BruteForce
- Exact Polynomial Algorithms
- Approximation

# Different Algorithms: BruteForce

We can take all possible combinations of the supports for all players and check if it has reached a mixed nash equilibrium.

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But,

Number of combinations:

$$= \prod_{i=1}^{k} \left(2^{M_i} - 1\right)$$

M = number of pure strategies available to a player

k = number of players in the game.

# Different Algorithms: BruteForce

But taking all the possible combinations will yield exponential time complexity !!

# Different Algorithms

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# Exact Polynomial Algorithms

## Two-player Zero-Sum Games

A situation where one player's gain is the other player's loss.
The total gains and losses of the players add up to **zero.**

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- Player 2 wins if the sum is **odd**.

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- Player 1 wins if the sum is **even**.
- Player 2 wins if the sum is **odd**.
- The winner gets **1** dollar from the loser.

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Rules:

- Players have to raise either one or two fingers.
- Player 1 wins if the sum is **even**.
- Player 2 wins if the sum is **odd**.
- The winner gets **1** dollar from the loser.
- The sum of net winnings is **zero**.

# Exact Polynomial Algorithms

## Two-player Zero-Sum Games

| Raising Finger Game | | Player 2 | |
|---|---|---|---|
| | | Strategy 1 | Strategy 2 |
| Player 1 | Strategy 1 | | |
| | Strategy 2 | | |

Table: Payoff Matrix

# Exact Polynomial Algorithms

## Two-player Zero-Sum Games

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| | | Strategy 1 | Strategy 2 |
| Player 1 | Strategy 1 | 1 | |
| | Strategy 2 | | |

Table: Payoff Matrix

# Exact Polynomial Algorithms

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Table: Payoff Matrix

Expected payoff of player 1 = Expected loss of player 2

Player 1 Maximizes payoff

Player 2 Minimizes loss

# Exact Polynomial Algorithms

**Two-player Zero-Sum Games**

For Two-Player-Zero-Sum games, finding a Nash equilibrium can often be done in **polynomial time** using *linear programming methods.*
Linear programs can be solved efficiently.

# Different Algorithms

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- Approximation Algorithms

## What is $\epsilon$-Nash equilibrium?

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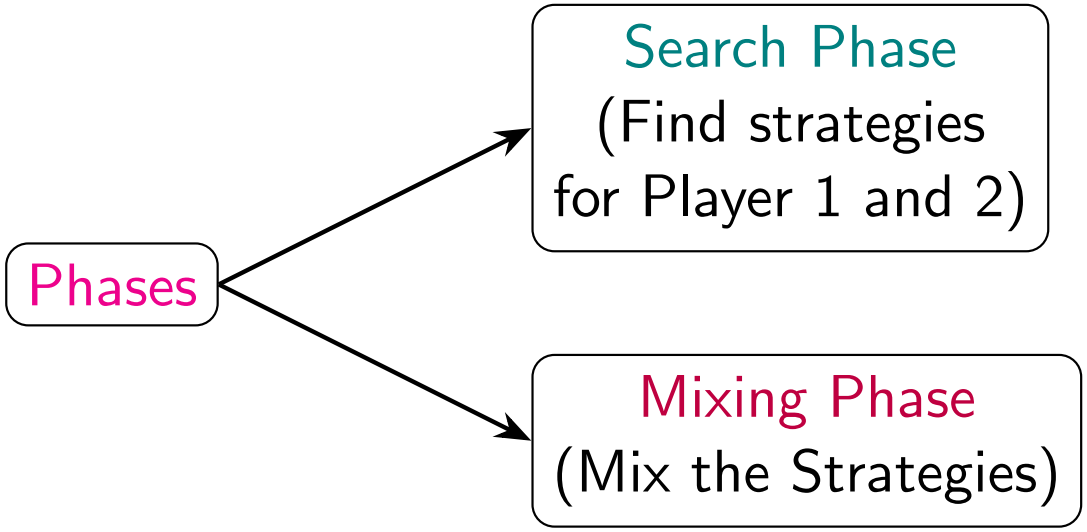
**Definition 1(Nash Equilibrium).**
A pair of strategies $(\hat{x}, \hat{y})$ is a Nash equilibrium for the bimatrix game $\Gamma = (A, B)$ if
(i) For every (mixed) strategy $x$ of the row player, $x^T A\hat{y} \leq \hat{x}^T A\hat{y}$ and
(ii) For every (mixed) strategy $y$ of the column player, $\hat{x}^T By \leq \hat{x}^T B\hat{y}$.

## What is $\epsilon$-Nash equilibrium?

**Definition 2($\epsilon$-Nash Equilibrium).**
For any $\varepsilon > 0$, a pair of strategies $(\hat{x}, \hat{y})$ is an $\varepsilon$-Nash equilibrium for the bimatrix game $\Gamma = (A, B)$ if
(i) For every (mixed) strategy $x$ of the row player,
$x^T A\hat{y} \leq \hat{x}^T A\hat{y} + \varepsilon$, and
(ii) For every (mixed) strategy $y$ of the column player,
$\hat{x}^T By \leq \hat{x}^T B\hat{y} + \varepsilon$.

# $\epsilon$ Nash Equilibrium



# Time Complexeties of $\epsilon$ -Nash Equilibrium

| Name | Bound proven by original papers | Bound computed from programs | Time complexity | Running time from programs (sec) |
|---|---|---|---|---|
| $\frac{3}{4}$ - Nash equilibrium for bimatrix game | 0.75 | 0.75 | $O(mn)$ | 1.39 |
| | | | | |
| | | | | |

Table: Table of Results

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| $\frac{1}{2}$ - Nash equilibrium for bimatrix game | 0.5 | 0.5 | $O(mn)$ | 0.06 |
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| $\frac{1}{2}$ - Nash equilibrium for bimatrix game | 0.5 | 0.5 | $O(mn)$ | 0.06 |
| $\frac{(3-\sqrt{5})}{2}$ - Nash equilibrium for bimatrix game | 0.381966 | 0.381966 | Polynomial time | 4.63 |

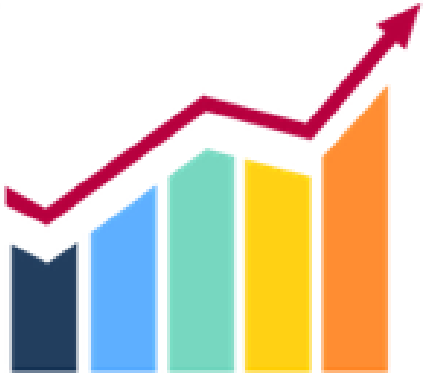
Table: Table of Results

# Applications

Figure: Applications of Nash Equilibrium

# Theoretical Application

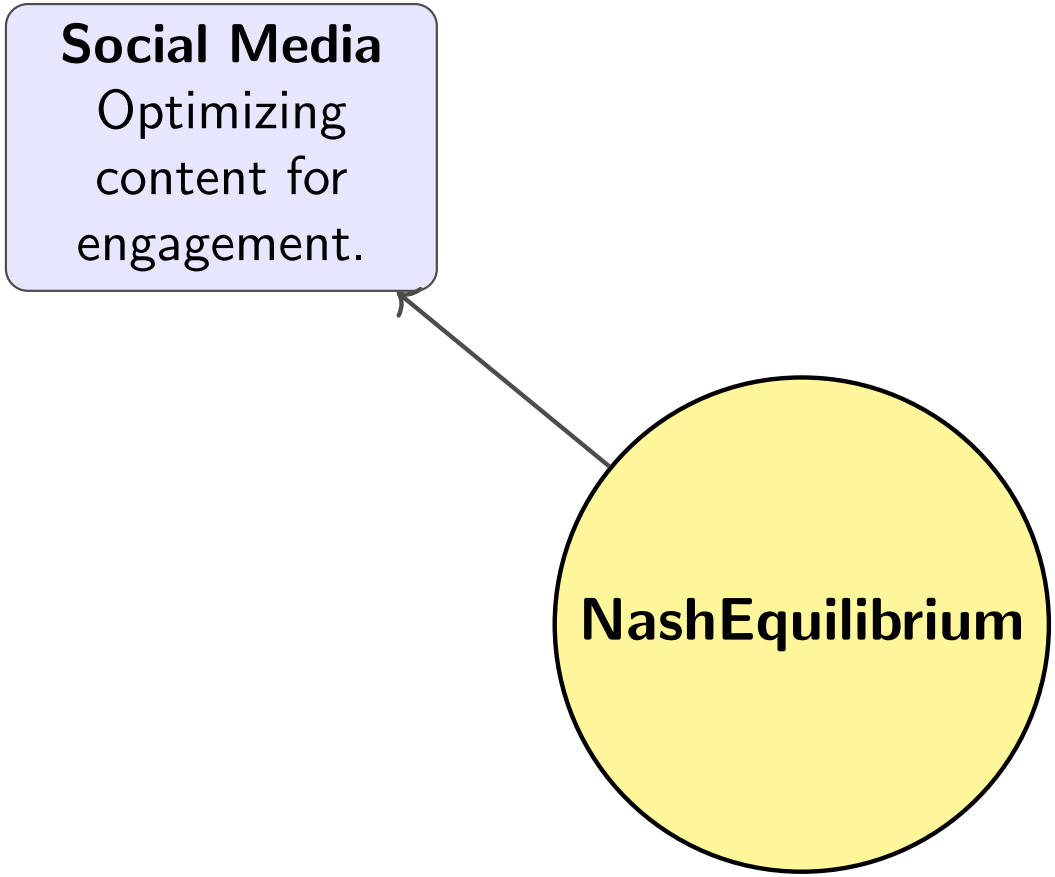
(a)
Game Theory Analysis
Nash Equilibrium is used to predict and analyze the outcomes of strategic games.



(b)
Economic Models
Nash Equilibrium is used to analyze oligopolistic markets, bidders' behaviors in auctions etc.

(c)
Biology and Evolution
Nash Equilibrium helps to understand competition and cooperation strategies among species populations.

# Applications of Nash Equilibrium



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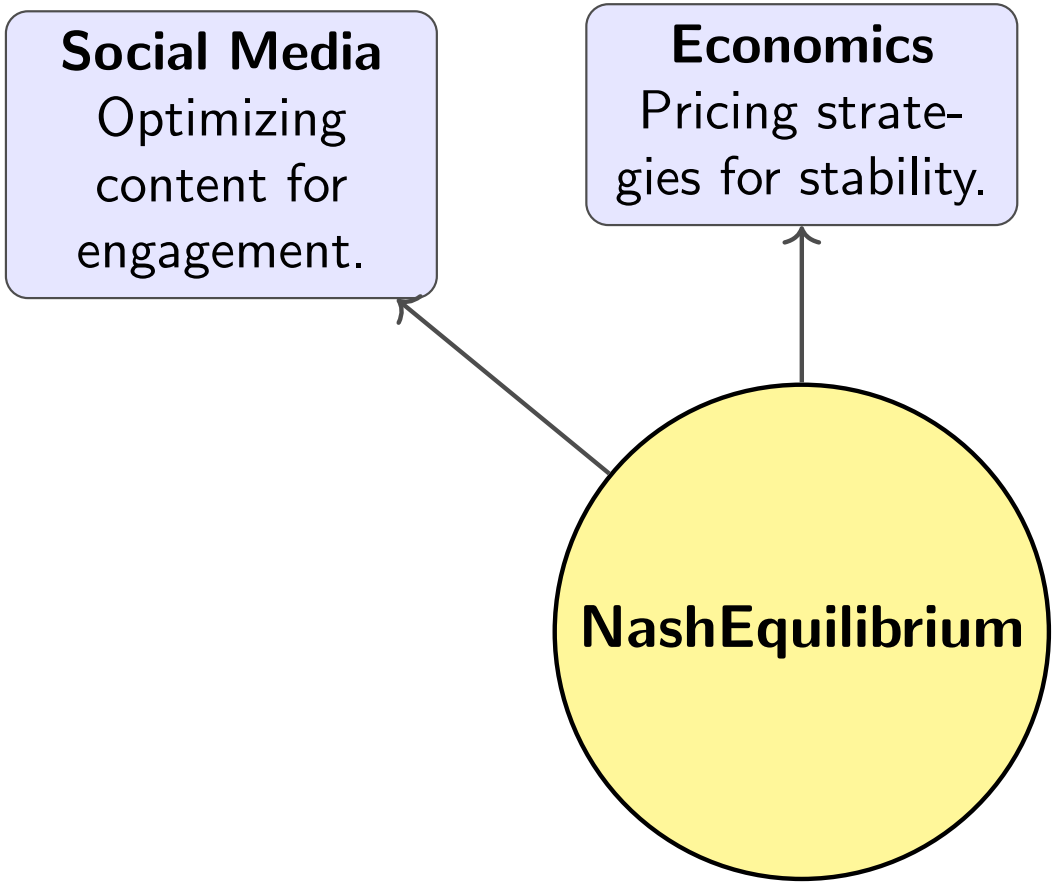

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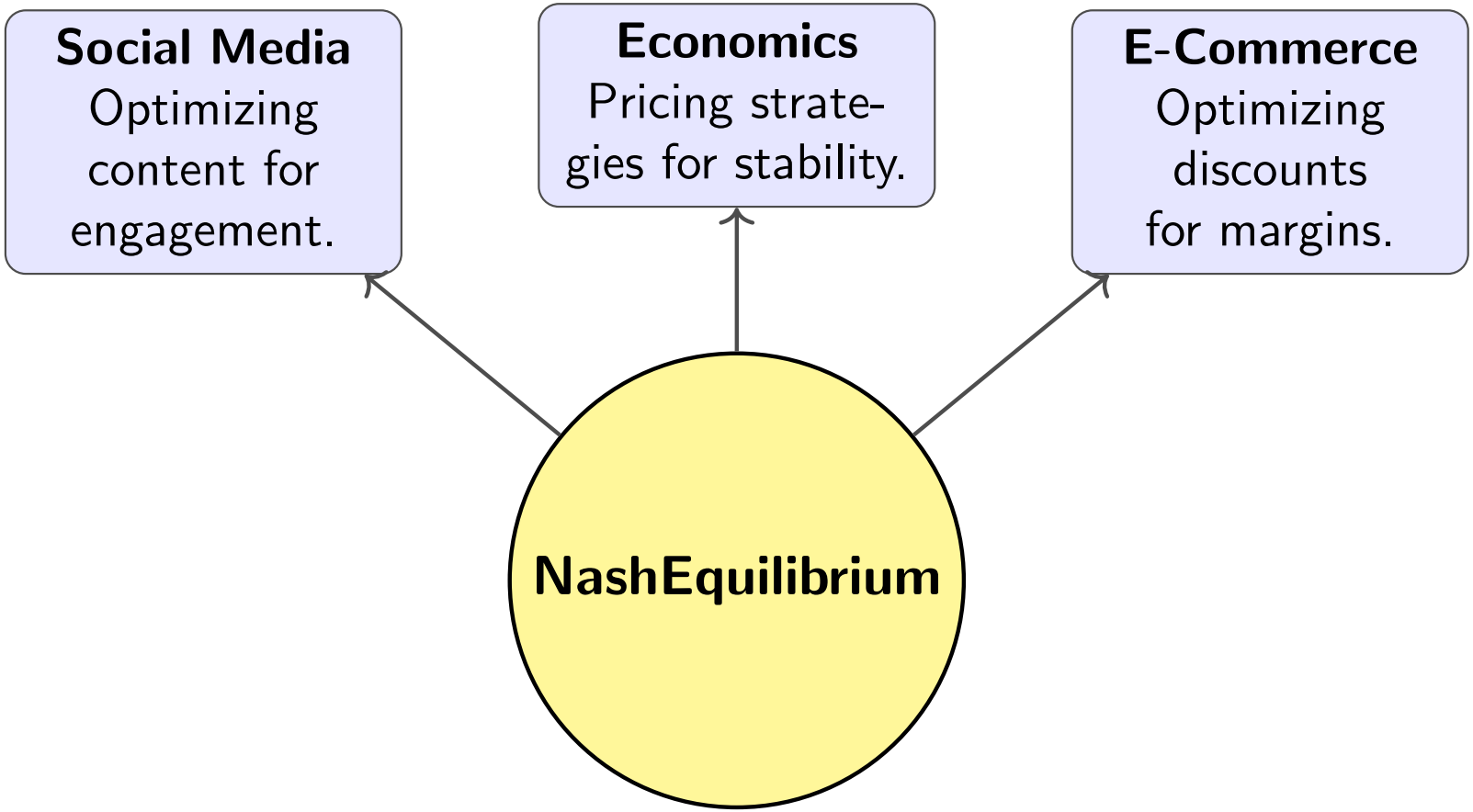

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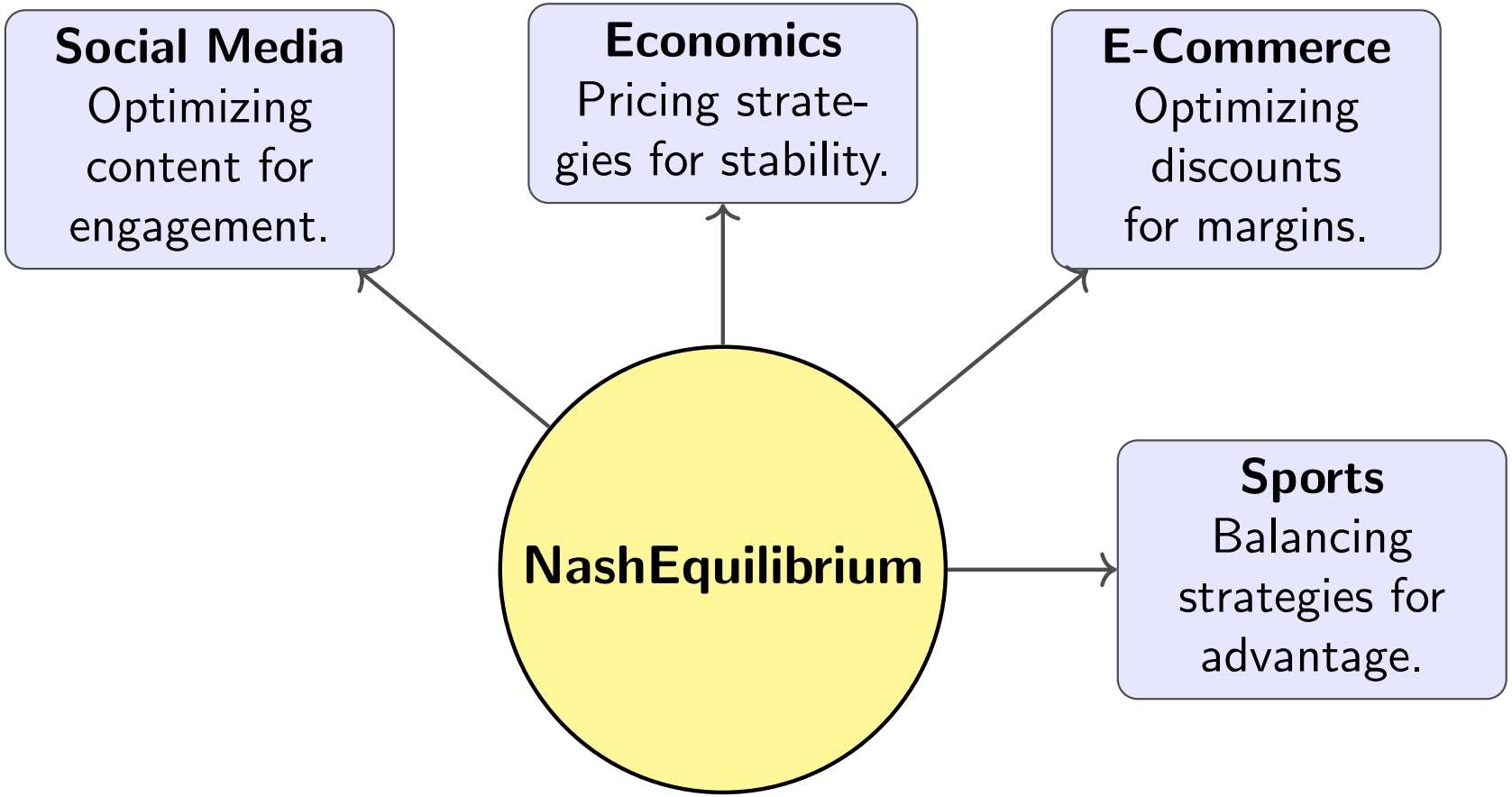

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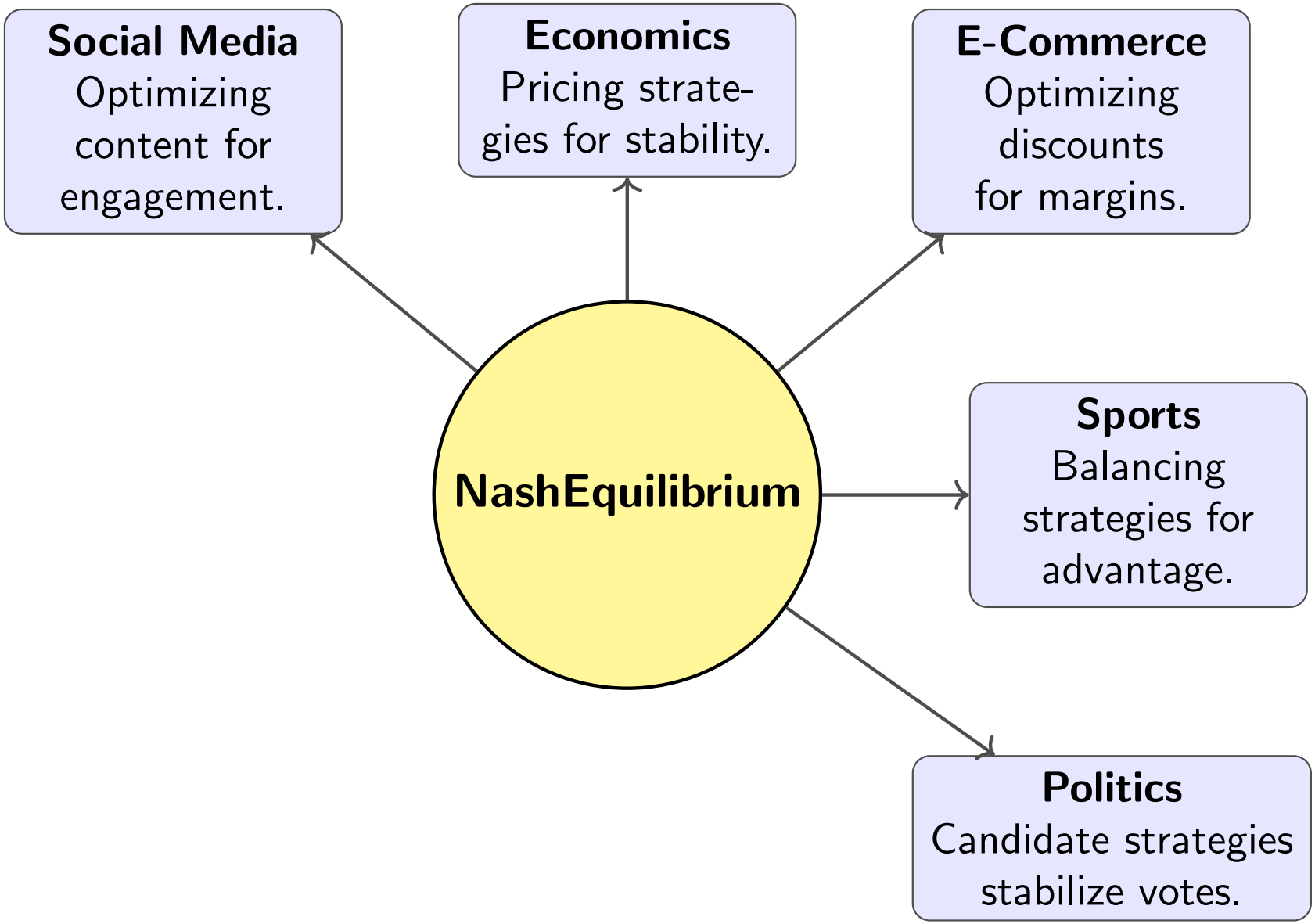

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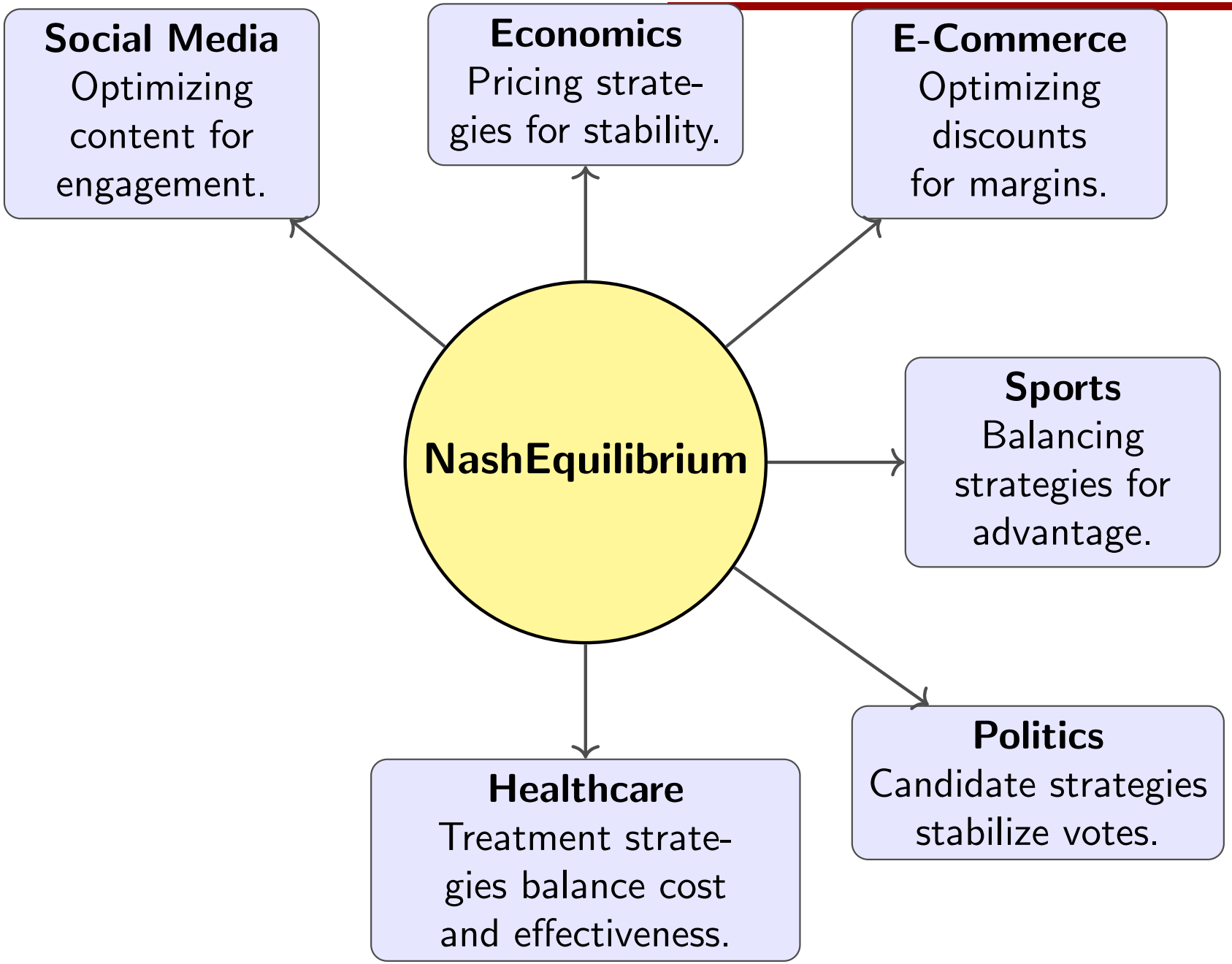

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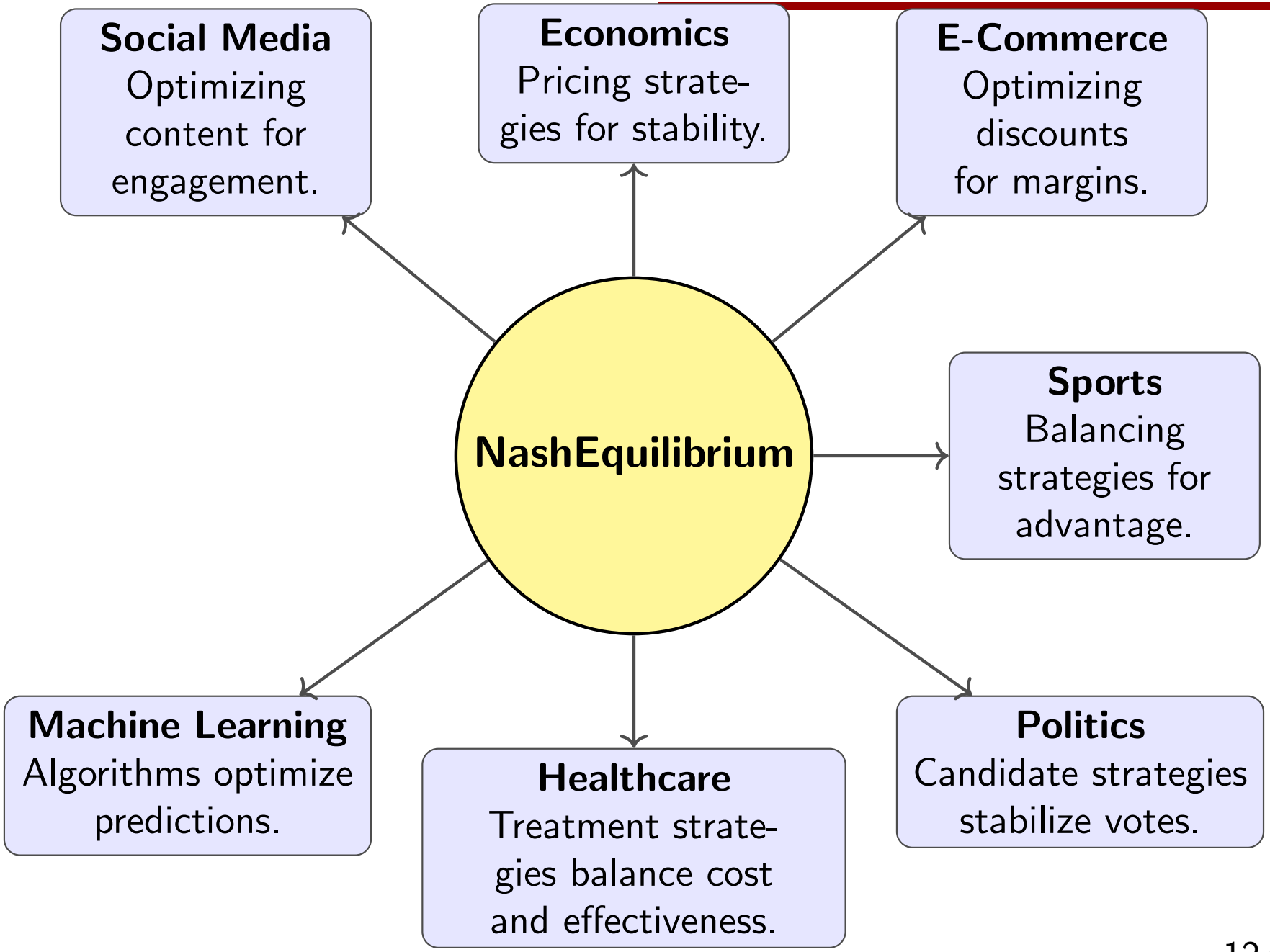

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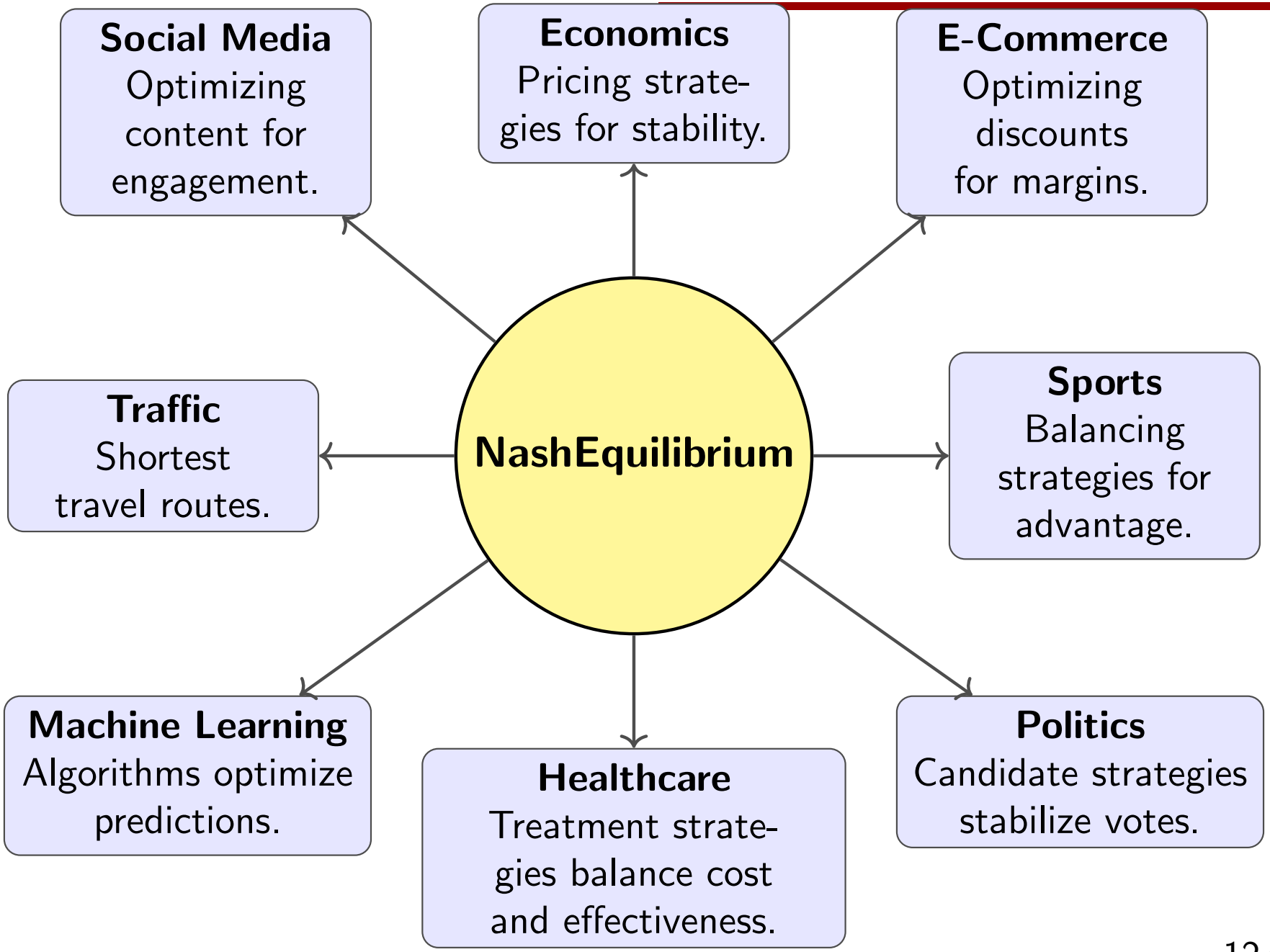
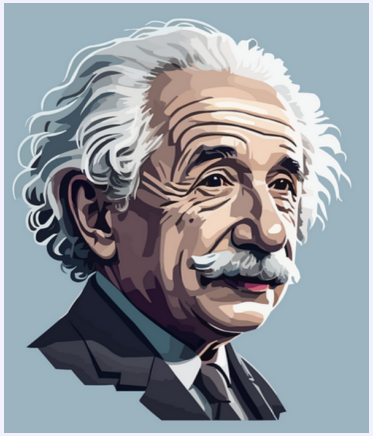


# Real Life Examples

## Quote by Albert Einstein



*"Life is just like a game,*
*First you have to learn rules of the game,*
*And then play it better than anyone else."*

**- Albert Einstein**

# Real Life Applications

**Sports:**

- Teams must decide on their strategies in competitive sports.

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- For example, football teams need to choose defensive, attacking, or neutral strategies in response to the opponents' strategies.

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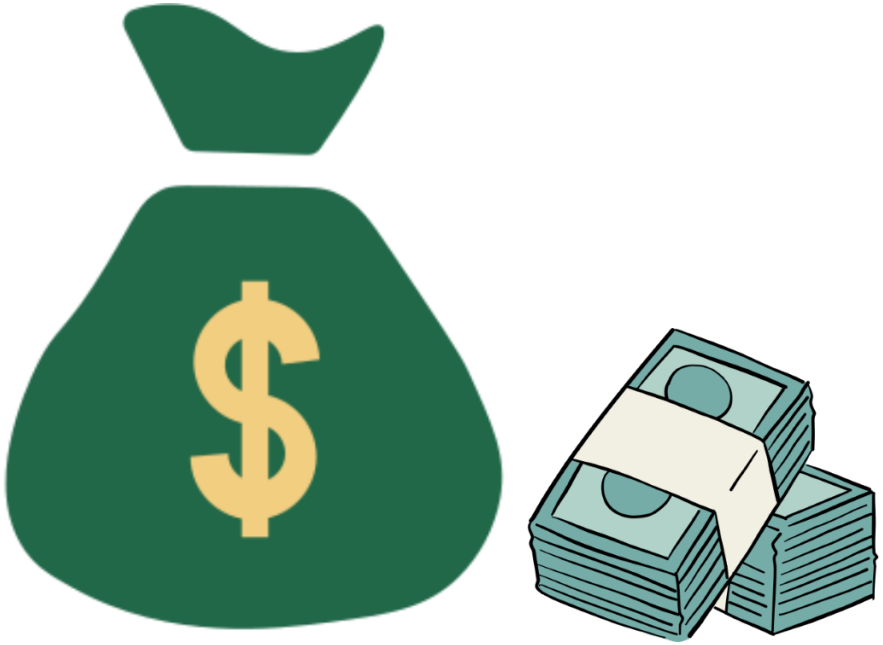
- For example, football teams need to choose defensive, attacking, or neutral strategies in response to the opponents' strategies.
- Too much defensive or too much attacking strategy may result in severe failure.

# Price Wars

- In highly competitive markets, companies often engage in price wars to gain market share and attract customers.



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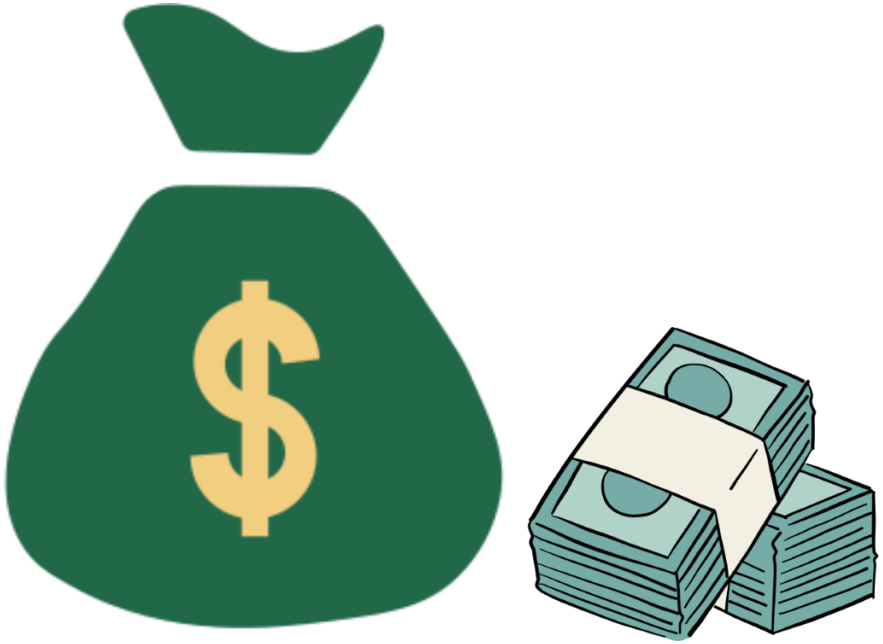
- Controlling the supply in the market allows companies to influence prices to rise.



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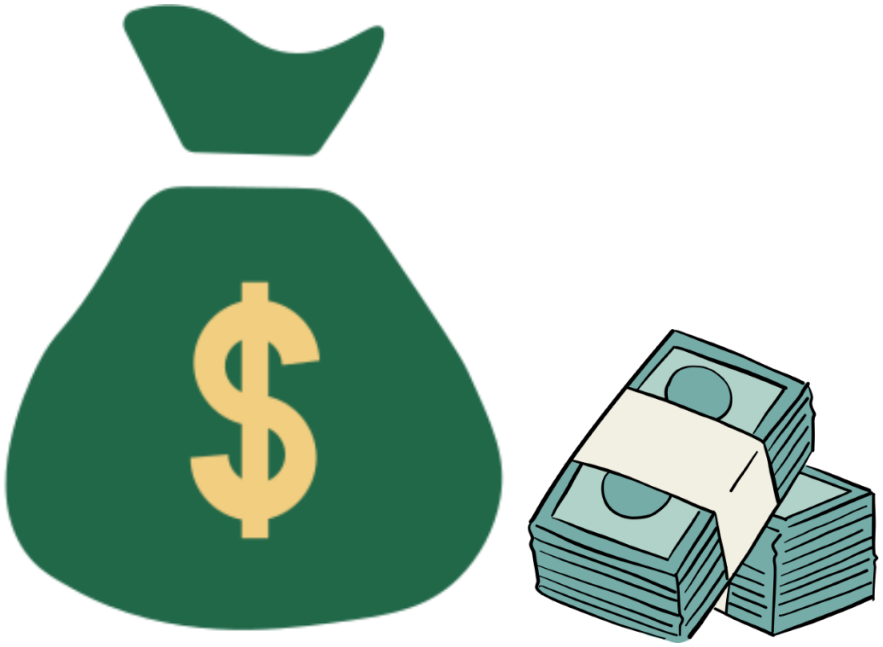
- Controlling the supply in the market allows companies to influence prices to rise.
- Competitors can attract more customers by cutting prices.



*THANK YOU*